

BAXZ 1313

Data Communications

Combined Tutorial Solutions

Full Worked Solutions — All Four Tutorials

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PART 1: SIGNALS

Lecture 4 — Full Worked Solutions

Question 1: Analog vs. Digital Signals

The table below compares analog and digital signals across the three dimensions requested.

Criterion	Analog signal	Digital signal
(a) Number of possible values	Infinite number of values; the signal level varies continuously over time (continuous range of amplitudes).	Finite number of values, typically just two discrete levels (0 and 1), though multilevel digital signals are also possible.
(b) Typical examples	Human voice, sound waves, old radio and TV broadcast signals, sine waves, temperature readings.	Computer data, square-wave pulses, ASCII-encoded text, signals on digital circuit buses.
(c) Usage in data communications	Used mainly for long-distance transmission over telephone lines and broadcast media; more susceptible to noise and attenuation.	Used in modern computer networks and LANs; more resistant to noise, supports error detection/correction, and is easier to regenerate.

Question 2: Period of a 5 kHz Signal

Given: frequency $f = 5 \text{ kHz} = 5000 \text{ Hz}$.

(a) **Period:**

$$T = 1/f = 1/5000 = 0.0002 \text{ s}$$

(b) **In microseconds:**

$$T = 0.0002 \text{ s} \times 10^6 = 200 \mu\text{s}$$

Question 3: Sine Wave Parameters

Given: $s(t) = 10 \sin(2\pi \cdot 1000t + \pi/3)$

Comparing with the general form $s(t) = A \sin(2\pi ft + \varphi)$:

(a) **Peak amplitude:** $A = 10$ (units, e.g. volts)

(b) **Frequency:** $f = 1000 \text{ Hz}$

(c) **Phase in degrees:**

$$\varphi = (\pi/3) \text{ rad} \times (180^\circ/\pi) = 60^\circ$$

Question 4: Phase Shift of 1/4 Cycle

(a) **Phase shift in degrees:**

$$1 \text{ full cycle} = 360^\circ \rightarrow 1/4 \text{ cycle} = 360^\circ/4 = 90^\circ$$

(b) Phase shift in radians:

$$90^\circ \times (\pi/180^\circ) = \pi/2 \text{ rad}$$

(c) What phase represents: Phase describes the position of the waveform relative to a fixed time reference ($t = 0$). It indicates how far the wave is shifted forward or backward in time, i.e. its horizontal displacement along the time axis.

Question 5: Composite Signal Bandwidth

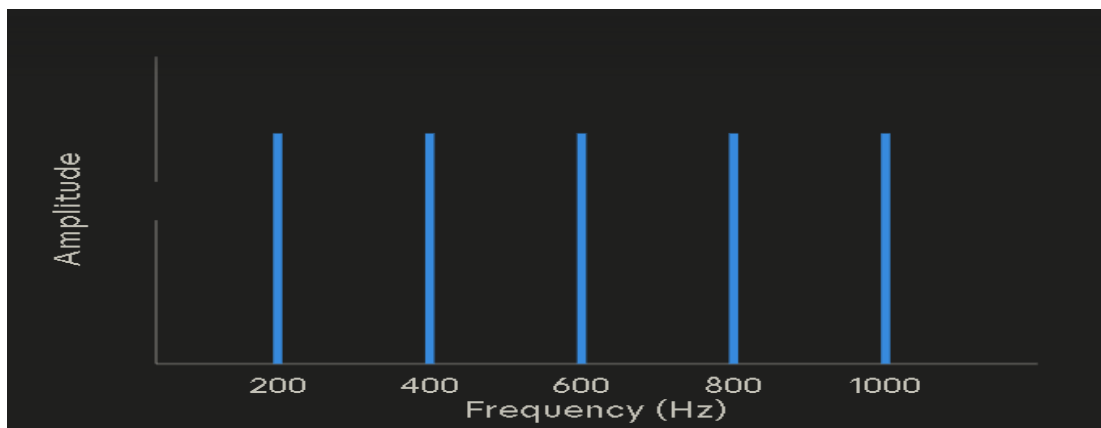
Given component frequencies: 200, 400, 600, 800, 1000 Hz.

(a) Highest and lowest frequency: Highest = 1000 Hz, Lowest = 200 Hz

(b) Bandwidth:

$$BW = f_{\text{highest}} - f_{\text{lowest}} = 1000 - 200 = 800 \text{ Hz}$$

(c) Frequency spectrum sketch: The spectrum consists of five discrete vertical lines (spikes) of equal amplitude, positioned at 200 Hz, 400 Hz, 600 Hz, 800 Hz, and 1000 Hz along the frequency axis:



Question 6: Bit Rate and Bit Interval

Given: bit rate = 4 Mbps = 4,000,000 bps.

(a) Bit interval:

$$\text{Bit interval} = 1 / \text{bit rate} = 1 / 4,000,000 = 0.00000025 \text{ s}$$

(b) In nanoseconds:

$$0.00000025 \text{ s} \times 10^9 = 250 \text{ ns}$$

(c) Relationship: Bit rate and bit interval are inversely related — as bit rate increases, bit interval decreases, and vice versa, following Bit Interval = 1 / Bit Rate.

Question 7: Channel Capacity: Shannon and Nyquist

Given: bandwidth $B = 2 \text{ MHz}$, $\text{SNR} = 31$.

(a) Shannon capacity:

$$C = B \log_2(1 + SNR) = 2,000,000 \times \log_2(1 + 31) = 2,000,000 \times \log_2(32)$$

$$\log_2(32) = 5 \rightarrow C = 2,000,000 \times 5 = 10,000,000 \text{ bps} = 10 \text{ Mbps}$$

(b) Nyquist bit rate (4 signal levels, L = 4):

$$C = 2 \times B \times \log_2(L) = 2 \times 2,000,000 \times \log_2(4) = 2 \times 2,000,000 \times 2$$

$$C = 8,000,000 \text{ bps} = 8 \text{ Mbps}$$

(c) Why Shannon capacity is the upper theoretical limit: Shannon's formula accounts for the presence of noise in the channel and gives the absolute maximum data rate that cannot be exceeded by any modulation or coding scheme, however sophisticated. Nyquist's formula, in contrast, assumes a noiseless channel and is only valid for an idealized case. Because real channels always contain noise, Shannon's capacity represents the true ceiling that practical systems can only approach, while actual achieved rates are usually lower due to real-world implementation constraints.

PART 2: ANALOG TRANSMISSION

Lecture 5 — Full Worked Solutions

Question 1: Digital-to-Analog Modulation

Digital-to-analog modulation is the process of changing one or more characteristics of an analog carrier signal — amplitude, frequency, or phase — to represent digital data.

Why modulation is required: Many transmission media (e.g. telephone lines, wireless channels) are designed to carry analog signals efficiently over long distances, or only support a specific frequency range. Sending raw digital signals directly would suffer severe attenuation and distortion. Modulation maps digital data onto an analog carrier suited to the channel's characteristics, enabling efficient long-distance transmission and allowing multiple signals to share the same medium (multiplexing).

Bit rate vs. baud rate: Bit rate is the number of bits transmitted per second (bps). Baud rate is the number of signal units (symbol changes) transmitted per second. If each signal unit carries more than one bit, baud rate is lower than bit rate; they are equal only when each signal unit represents exactly one bit.

Question 2: Baud Rate and Bit Rate

Given: 3 bits per signal unit, 6000 signal units transmitted per second.

(a) **Baud rate:** = 6000 baud (the number of signal units per second, given directly).

(b) **Bit rate:**

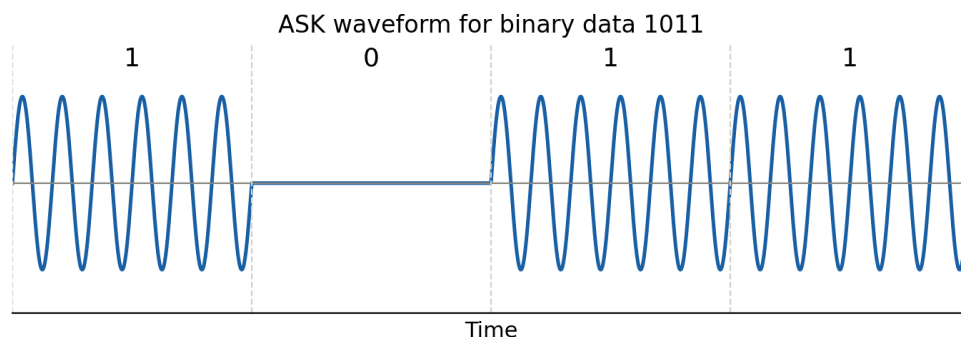
$$\text{Bit Rate} = \text{Baud Rate} \times \text{bits per signal unit} = 6000 \times 3 = 18,000 \text{ bps}$$

(c) **Relationship:** Bit rate = Baud rate $\times$ (number of bits per signal element). Baud rate is always less than or equal to bit rate, and they are equal only when one bit is carried per signal unit.

Question 3: Amplitude Shift Keying (ASK)

(a) **How ASK represents binary data:** Binary 1 is represented by transmitting the carrier at full (high) amplitude; binary 0 is represented by zero or much lower amplitude, while frequency and phase remain constant.

(b) **Waveform for binary data 1011:**



(c) Advantage / disadvantage: Advantage: simple to implement and demodulate, low cost. Disadvantage: highly susceptible to noise and sudden gain changes, since the information is carried in amplitude, which noise easily corrupts.

Question 4: Full-Duplex ASK Channel Allocation

Given: total bandwidth from 2000 Hz to 12,000 Hz (total span = 10,000 Hz).

(a) Bandwidth per direction:

$$\text{Bandwidth per direction} = 10,000 \text{ Hz} / 2 = 5000 \text{ Hz}$$

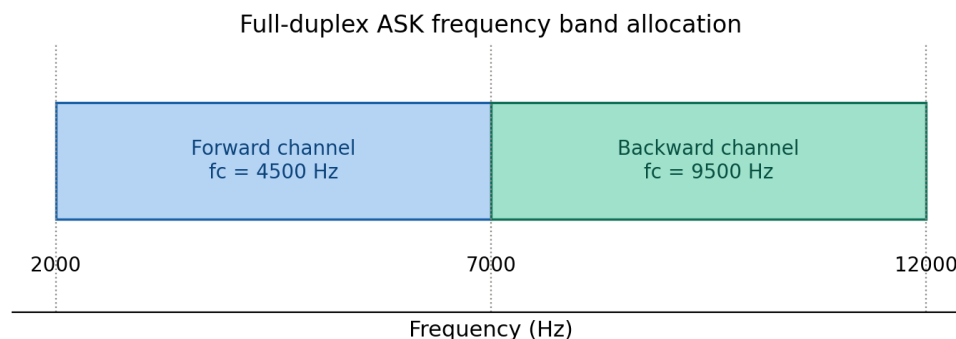
Forward channel: 2000–7000 Hz. Backward channel: 7000–12,000 Hz (each 5000 Hz wide).

(b) Carrier frequency for each direction: The carrier frequency is placed at the center of each sub-band.

$$\text{Forward carrier} = (2000 + 7000) / 2 = 4500 \text{ Hz}$$

$$\text{Backward carrier} = (7000 + 12000) / 2 = 9500 \text{ Hz}$$

(c) Frequency bands illustration:

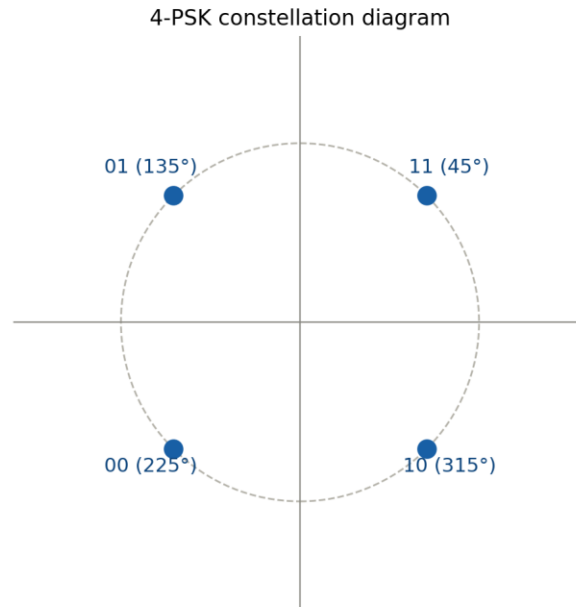


Question 5: Phase Shift Keying (PSK)

PSK represents digital data by varying the phase of the carrier signal, while amplitude and frequency remain constant.

(a) 2-PSK (BPSK): Binary 0 is represented by one phase (e.g. 0°), and binary 1 by a phase shifted 180° from it. Only two phase states are used, each carrying 1 bit.

(b) 4-PSK constellation diagram:



(c) Bits per signal unit in 4-PSK: Each of the four phase points carries 2 bits per signal unit, since 4 distinct phase states represent $2^2 = 4$ possible bit combinations.

Question 6: 16-QAM

(a) Bits per signal unit:

$$\log_2(16) = 4 \text{ bits per signal unit}$$

(b) Bit rate (baud rate = 2400 baud):

$$\text{Bit Rate} = \text{Baud Rate} \times \text{bits per signal unit} = 2400 \times 4 = 9600 \text{ bps}$$

(c) Why QAM gives higher data rates: QAM varies both amplitude and phase simultaneously, producing far more distinguishable signal states than ASK (amplitude only) or PSK (phase only) for the same bandwidth. Because more bits can be packed into each signal unit, QAM achieves a higher bit rate than ASK or PSK alone for the same baud rate / bandwidth.

Question 7: AM and FM Bandwidth

Given: audio signal bandwidth $B = 5 \text{ kHz}$.

(a) AM bandwidth:

$$BW(AM) = 2 \times B = 2 \times 5 \text{ kHz} = 10 \text{ kHz}$$

(b) FM bandwidth: Using the standard broadcast approximation that FM bandwidth is about 10 times the audio bandwidth:

$$BW(FM) = 10 \times B = 10 \times 5 \text{ kHz} = 50 \text{ kHz}$$

(c) Why FM needs more bandwidth than AM: FM produces multiple sidebands due to frequency deviation and the modulation index, spreading signal energy across a much wider frequency range. AM produces only two simple sidebands directly proportional to the message

bandwidth, while FM's bandwidth depends on both the message bandwidth and the (typically much larger) frequency deviation.

PART 3: DIGITAL TRANSMISSION

Lecture 5(ii) — Full Worked Solutions

Question 1: Line Coding

Line coding is the process of converting digital data (bits 0 and 1) into a digital signal suitable for transmission over a communication medium.

Two important characteristics for reliable transmission:

- 1. Synchronization:** Line coding should include a mechanism that helps the receiver accurately identify the start and end of each bit, especially during long runs of consecutive 0s or 1s, so the receiver does not lose clock synchronization with the transmitter.
- 2. DC Component (DC balance):** It is preferable for the average value of the signal to be zero (no DC component), since a DC component causes problems when passing through transformers or lines that cannot carry very low frequencies, and increases unnecessary power consumption.

Question 2: Signal Level vs. Data Level

(a) Difference: Data level is the number of possible values in the data itself (e.g. binary data has only 2 data levels: 0 and 1). Signal level is the number of distinct values actually used in the transmitted signal, which may be greater than the number of data levels depending on the encoding scheme.

(b) How a 2-data-level signal can use more than 2 signal levels: This occurs when an encoding scheme (such as Manchester encoding) introduces extra transitions or signal levels to represent a single bit, in order to improve synchronization or remove the DC component, even though the original data is purely binary (0 and 1).

(c) Advantage of using more signal levels: Using a greater number of signal levels allows more than one bit to be carried per signal unit, which increases the bit rate without needing to increase the baud rate or the channel bandwidth.

Question 3: Bits per Signal Unit (4 Signal Levels)

Given: 4 signal levels ($L = 4$), pulse rate = 3000 pulses per second.

(a) Bits per signal unit:

$$\text{bits per signal unit} = \log_2(L) = \log_2(4) = 2 \text{ bits}$$

(b) Bit rate:

$$\text{Bit Rate} = \text{Pulse Rate} \times \log_2(L) = 3000 \times 2 = 6000 \text{ bps}$$

(c) Formula used:

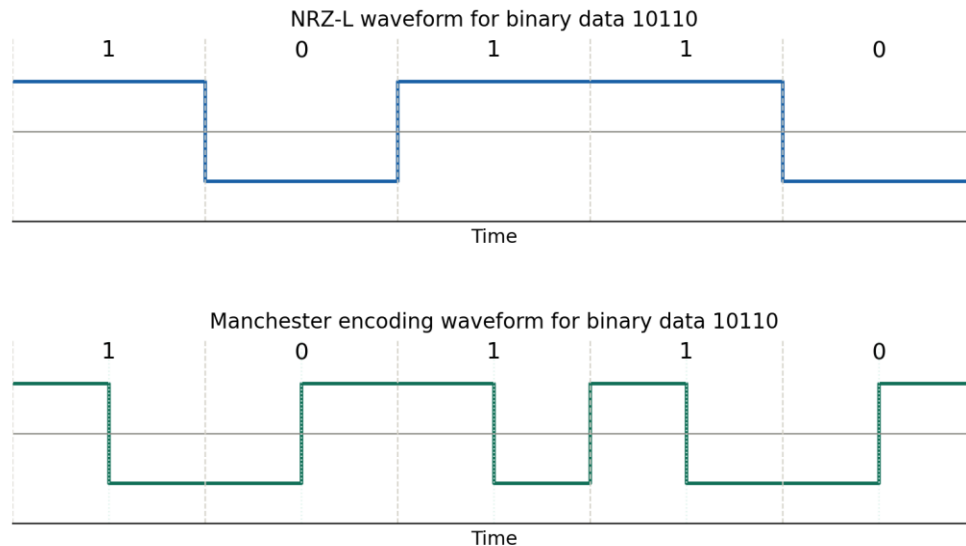
$$\text{Bit Rate} = \text{Baud Rate} \times \log_2(L), \text{ where } L \text{ is the number of signal levels}$$

Question 4: NRZ-L and Manchester Encoding

(a) NRZ-L (Non-Return-to-Zero Level): Binary 1 is represented by one constant signal level (e.g. high/positive voltage), and binary 0 by a different level (e.g. low/negative voltage). The signal does not return to zero between bits, and a transition occurs only when the bit value changes.

(b) Manchester encoding: Combines data and clock timing information in a single signal. Each bit is represented by a transition in the middle of the bit interval: a high-to-low transition represents one bit value and a low-to-high transition represents the other (the exact convention depends on the standard used, e.g. IEEE 802.3 or the G.E. Thomas convention).

Waveforms for binary data 10110:



The two waveforms above show the same bit sequence (10110) encoded differently: in NRZ-L the signal stays constant for the whole bit duration and changes only when the bit value changes, while in Manchester encoding a transition always occurs at the mid-point of every bit, regardless of its value.

Question 5: DC Component and Self-Synchronization

(a) What is a DC component: The DC component is the zero-frequency component of a signal, i.e. its non-zero average value over a long period of time. It arises when a signal does not spend an equal amount of time at its high and low levels.

(b) Why a DC component is undesirable: Transformers used in some transmission lines cannot pass a DC component, causing signal loss or distortion. It also wastes power without carrying any information, and can cause problems in detection at the receiver.

(c) How Manchester / Differential Manchester encoding helps self-synchronization: Both schemes guarantee at least one transition in the middle of every bit interval, regardless of the bit's value. This regular transition allows the receiver to extract the clock signal directly from the received signal itself, without needing a separate timing signal, keeping the receiver synchronized with the transmitter even during long runs of 0s or 1s.

Question 6: Nyquist Sampling Theorem

Given: analog signal bandwidth from 500 Hz to 5500 Hz.

(a) Highest frequency:

$$f_{\max} = 5500 \text{ Hz}$$

(b) Minimum sampling rate (Nyquist theorem):

$$f_s = 2 \times f_{\max} = 2 \times 5500 = 11,000 \text{ samples/second}$$

(c) Why undersampling causes distortion: When the sampling rate is lower than twice the highest frequency in the signal (the Nyquist rate), a phenomenon called aliasing occurs: samples of different frequencies become indistinguishable from each other, so the original signal cannot be correctly reconstructed from the samples. A false (alias) frequency appears, completely different from the true frequency present in the original signal.

Question 7: PCM and Bits per Sample

Given: 10 quantization levels required, sampling rate = 8000 samples per second.

(a) Minimum number of bits per sample:

$$n = \log_2(10) \approx 3.32 \rightarrow \text{rounded up} \rightarrow n = 4 \text{ bits}$$

($2^3 = 8$ levels is not enough; $2^4 = 16$ levels is sufficient to represent 10 levels.)

(b) Bit rate:

$$\text{Bit Rate} = \text{Sampling Rate} \times \text{bits per sample} = 8000 \times 4 = 32,000 \text{ bps}$$

(c) Role of PCM (Pulse Code Modulation) in digital transmission: PCM is a technique for converting an analog signal into a digital signal through three main steps: Sampling, Quantization, and Encoding. PCM allows analog signals (such as voice) to be carried over digital systems, gaining the benefits of digital transmission such as improved signal quality, the ability to detect and correct errors, and easy signal regeneration over long distances.

PART 4: ERROR DETECTION AND CORRECTION

Lecture 08 — Full Worked Solutions

Question 1: Single-bit Error and Burst Error

(a) Definitions:

Single-bit error: An error in which only one bit in a transmitted data unit is changed (from 0 to 1 or from 1 to 0), while all other bits remain correct.

Burst error: An error in which two or more consecutive bits in a data unit are corrupted. Not every bit between the first and last corrupted bit needs to be wrong, but the first and last corrupted bits define the length of the burst (burst length).

(b) Why burst errors are more common than single-bit errors: In most transmission media, the bit rate is very high. If noise or interference occurs for even a short duration, it affects several consecutive bits rather than just one, because the duration of the noise is typically longer than the time needed to transmit a single bit (bit duration).

(c) Practical example of burst errors: In wireless communication, sudden electrical interference (e.g. a nearby appliance switching on) or signal fading affects a whole segment of the signal over a short time period, corrupting several consecutive bits at once.

Question 2: Error Detection vs. Error Correction

(a) Difference:

Error detection: Only determines whether an error has occurred in the received data, without identifying exactly which bit is wrong or fixing it. Once an error is detected, the receiver requests retransmission from the sender.

Error correction: Goes beyond detection by precisely identifying the location of the corrupted bit(s) and automatically correcting them, without needing retransmission.

(b) Why data-link layer protocols prefer detection over correction: Error correction requires adding a much larger number of redundant bits compared to detection alone, increasing both computational overhead and bandwidth overhead. Since retransmission is usually fast and simple in modern networks, it is more efficient to use error detection only (e.g. CRC) and request retransmission when an error is found, rather than bearing the cost and complexity of correction mechanisms every time.

Question 3: Even Parity Bit Calculation

For even parity, the parity bit is chosen so that the total number of 1s (data bits + parity bit) is even.

Data unit	Number of 1s	Parity bit	Final transmitted data
(a) 1010110	4 (even)	0	10101100
(b) 1101001	4 (even)	0	11010010
(c) 0001111	4 (even)	0	00011110

Question 4: 2D Even Parity

Given data block:

1010100 0011100 1101110 1110011

Row parity: each row's parity bit is chosen so the total number of 1s in that row (including the parity bit) is even. Column parity: each column's parity bit is chosen so the total number of 1s in that column (including the parity bit) is even.

1	0	1	0	1	0	0	1
0	0	1	1	1	0	0	1
1	1	0	1	1	1	0	1
1	1	1	0	0	1	1	1
1	0	1	0	1	0	1	

The rightmost column shows the row parity bits, and the bottom row shows the column parity bits, each computed so that the total count of 1s (including the parity bit) in that row/column is even.

Question 5: Cyclic Redundancy Check (CRC)

(a) Principle of CRC: CRC treats the transmitted bit sequence as a polynomial. The sender performs a modulo-2 division of the dataword by a pre-agreed fixed divisor (the generator polynomial). The remainder of this division, called the CRC, is appended to the original data to form the transmitted codeword. At the receiving end, the receiver performs the same modulo-2 division on the entire received codeword; if the remainder is zero, the data is considered error-free, and if it is non-zero, an error has occurred during transmission.

(b) Two advantages of CRC over simple parity check:

- 1. Higher detection capability:** CRC can detect a much wider range of error patterns, including all single-bit errors, most double-bit errors, all errors with an odd number of corrupted bits, and importantly, most burst errors, which simple parity check often fails to detect.
- 2. Good efficiency for the protection provided:** CRC adds a relatively small number of redundant bits compared to the high level of error-detection capability it provides, giving a good balance between overhead and protection.

Question 6: CRC Calculation by Modulo-2 Division

Given: dataword = 100100, divisor = 1101 (4 bits, so the CRC has 3 bits).

Step 1 — Append zeros: Append (divisor length – 1) = 3 zeros to the dataword:

100100 000 → 100100000

(a) Modulo-2 division (XOR at each step):

Step	Active 4 bits	XOR with divisor (1101)
1	1001	$1001 \oplus 1101 = 0100$

2	1000	$1000 \oplus 1101 = 0101$
3	1010	$1010 \oplus 1101 = 0111$
4	1110	$1110 \oplus 1101 = 0011$
5	0110	leading bit 0 $\rightarrow$ no XOR $\rightarrow$ 0110
6	1100	$1100 \oplus 1101 = 0001$

(Each step uses XOR with the divisor when the leading bit of the active window is 1, and XOR with 0000 — i.e. no change — when it is 0; the leftmost bit is then dropped and the next bit of the dividend is brought down, exactly as in long division.)

(b) CRC remainder:

Remainder = 001

(c) Final transmitted codeword:

Codeword = Dataword + Remainder = 100100 + 001 = 100100001